Review

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Chapter 1: Linear Space and Mapping

Section 1.1 Basic abstract algebra

Algebraic systems: groups,rings and fields

Section 1.2 Linear space

definition: a set with addition “+” closed and
scala multiplication associative and distributive
linear dependence, linear combination

Section 1.3 Basis of Linear space

Section 1.4 Linear subspace

Section 1.5 Linear transformations

Outline-Linear Space and Mapping

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Section 1.1 Basic abstract algebra
Section 1.2 Linear space

Section 1.3 Basis of Linear space

What is it? Maximal linearly independent set
Dimension of linear space? Number of vectors in the basis
Change of basis, transition matrices,

Section 1.4 Linear subspace

What is a linear subspace?
Operations between subspaces: sum, intersection, direct sum,

Section 1.5 Linear transformations

What is it? Linear mapping from one linear space to another
How to represent? Matrix representation
Properties, range and kernel

1.3 Basis of a linear space

•1.3.1 The notion of a basis

Definition 1.15 A finite set of vectors $\boldsymbol{\alpha}_1$, $\boldsymbol{\alpha}_2$, ..., $\boldsymbol{\alpha}_n$ in a linear space $\mathcal{V}$ over the field $\mathcal{F}$ is referred to as a basis of $\mathcal{V}$ if they are linearly independent and every vector $\boldsymbol{\alpha}$ in $\mathcal{V}$ is a linear combination of its elements, that is

$$\boldsymbol{\alpha} = k_{\alpha,1}\boldsymbol{\alpha}_1 + k_{\alpha,2}\boldsymbol{\alpha}_2 + \cdots + k_{\alpha,n}\boldsymbol{\alpha}_n$$

$$= [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \ldots \quad \boldsymbol{\alpha}_n] \begin{bmatrix} k_{\alpha,1} \\ k_{\alpha,2} \\ \vdots \\ k_{\alpha,n} \end{bmatrix}$$

for all $\boldsymbol{\alpha} \in \mathcal{V}$. The n-dimensional column vector $\boldsymbol{k}_{\boldsymbol{\alpha}} = [k_{\alpha,1} k_{\alpha,2} \ldots k_{\alpha,n}]^T$ is called the representation of $\boldsymbol{\alpha}$ with respect to the basis $\boldsymbol{\alpha}_1$, $\boldsymbol{\alpha}_2$, ..., $\boldsymbol{\alpha}_n$. $\mathcal{V}$ is called an n-dimensional linear space, denoted $\dim \mathcal{V} = n$.

Denote by $[\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \ldots \quad \boldsymbol{\alpha}_n]$ the basis of a linear space of dimension n.

1.3 Basis of a linear space

- Some explanations

A basis is also called a maximal linearly independent set of the space $\mathcal{V}$.

The $\dim \mathcal{V}$ is the number of vectors in any maximal linearly independent set of $\mathcal{V}$. In this sense, we say also that the rank of an n-dimensional space is n, denoted $\operatorname{rank} \mathcal{V} = n$

1.3 Basis of a linear space

- *An example*

Exp: $\mathcal{R}_{n-1}[\lambda]$ is an n-dimensional linear space.

Proof: Via the Exp 1.4 on Pg 8, the set of all polynomials with degree not exceeding $n-1$ and coefficiencts in $\mathcal{R}$ is a linear space. It is clear that the vectors 1, λ, λ^2, ..., λ^{n-1} are all in $\mathcal{R}_{n-1}[\lambda]$ and every $f(\lambda) \in \mathcal{R}_{n-1}[\lambda]$ can be represented as $f(\lambda) = a_0 + a_1\lambda + \cdots + a_{n-1}\lambda^{n-1}$, i.e. $f(\lambda)$ is a linear combination of these vectors. It suffices then to show that the vectors 1, λ, λ^2, ..., λ^{n-1} in $\mathcal{R}_{n-1}[\lambda]$ are linearly independent.

1.3 Basis of a linear space

- Example-continued

Suppose that there exist k_1, k_2, ..., k_{n-1}, all in $\mathcal{R}$ such that

$$k_0 \cdot 1 + k_1\lambda + \cdots k_{n-1}\lambda^{n-1} = 0\,.$$

Comparing the coefficients of the both sides we conclude that $k_i = 0$ for all $i = 0, 1, 2, \ldots, n$, which implies that 1, λ, λ^2, ..., λ^{n-1} are linearly independent. Thus $\dim \mathcal{R}_{n-1}[\lambda] = n$.

1.3 Basis of a linear space

- Example-continued

It can be readily verified that 1, $\lambda - a$, $(\lambda - a)^2$, ..., $(\lambda - a)^{n-1}$ can be a basis of $\mathcal{R}_{n-1}[\lambda]$ as well, since every polynomial $f(\lambda)$ with degree no more than $n-1$ can be expanded to the Taylor series

$$f(\lambda) = f(a) + f'(a)(\lambda - a) + \frac{f''(a)}{2!}(\lambda - a)^2 + \cdots + \frac{f^{(n-1)}(a)}{(n-1)!}(\lambda - a)^{n-1}$$

at $\lambda = a$. Thus the representation of $f(\lambda)$ with respect to this basis is

$$f(a), \quad f'(a), \quad \frac{f''(a)}{2!}, \quad \cdots, \quad \frac{f^{(n-1)}(a)}{(n-1)!}.$$

1.3 Basis of a linear space

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- 1.3.2 Change of Basis and Transition Matrices
- Basis is not unique, what is relation of different bases and the associated representation of a vector?

Let $[\boldsymbol{\alpha}_1\ \boldsymbol{\alpha}_2\ \ldots\ \boldsymbol{\alpha}_n]$ and $[\boldsymbol{\beta}_1\ \boldsymbol{\beta}_2\ \ldots\ \boldsymbol{\beta}_n]$ be two different bases of $\mathcal{V}$. Since $[\boldsymbol{\alpha}_1\ \ \boldsymbol{\alpha}_2\ \ \ldots\ \ \boldsymbol{\alpha}_n]$ form a basis of $\mathcal{V}$, $\boldsymbol{\beta}_i$ can be represented as

$$\boldsymbol{\beta}_i = p_{1i}\boldsymbol{\alpha}_1 + p_{2i}\boldsymbol{\alpha}_2 + \cdots + p_{ni}\boldsymbol{\alpha}_n = [\boldsymbol{\alpha}_1\ \ \boldsymbol{\alpha}_2\ \ \ldots\ \ \boldsymbol{\alpha}_n] \begin{bmatrix} p_{1i} \\ p_{2i} \\ \vdots \\ p_{ni} \end{bmatrix},$$

where $i = 1, 2, \ldots, n$. The n equations can be represented by a matrix equation as follows

$$[\boldsymbol{\beta}_1\ \ \boldsymbol{\beta}_2\ \ \ldots\ \ \boldsymbol{\beta}_n] = [\boldsymbol{\alpha}_1\ \ \boldsymbol{\alpha}_2\ \ \ldots\ \ \boldsymbol{\alpha}_n] \underbrace{\begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{bmatrix}}_{\triangleq P}$$

The $n \times n$ matrix P is called the transition matrix from the basis $[\boldsymbol{\alpha}_1\ \ \boldsymbol{\alpha}_2\ \ \ldots\ \ \boldsymbol{\alpha}_n]$ to the basis $[\boldsymbol{\beta}_1\ \ \boldsymbol{\beta}_2\ \ \ldots\ \ \boldsymbol{\beta}_n]$.

1.3 Basis of a linear space

- An theorem related to the transition matrix

Theorem 1.4 A transition matrix is invertible.

- The relationship between representation of a vector with respect to different bases?

Let ξ be any vector in $\mathcal{V}$, having the representations $\boldsymbol{x} = [x_1 \quad x_2 \quad \dots \quad x_n]^{\mathsf{T}}$ and $\boldsymbol{y} = [y_1 \quad y_2 \quad \dots \quad y_n]^{\mathsf{T}}$ with respect to the bases $\boldsymbol{\alpha}_1$, $\boldsymbol{\alpha}_2$, ..., $\boldsymbol{\alpha}_n$ and $\boldsymbol{\beta}_1$, $\boldsymbol{\beta}_2$, ..., $\boldsymbol{\beta}_n$ respectively, that is

$$\begin{aligned} \boldsymbol{\xi} &= [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n]\,\boldsymbol{x}, \\ \boldsymbol{\xi} &= [\boldsymbol{\beta}_1 \quad \boldsymbol{\beta}_2 \quad \dots \quad \boldsymbol{\beta}_n]\,\boldsymbol{y}, \end{aligned}$$

then

$$\boldsymbol{y} = P^{-1}\boldsymbol{x}$$

1.4 Linear subspaces

- Notation of linear subspaces

Definition 1.16 Let $\mathcal{W}$ be a subset of an n-dimensional linear space $\mathcal{V}$ over the field $\mathcal{F}$. If
1. $\boldsymbol{\alpha} + \boldsymbol{\beta} \in \mathcal{W}$ $\forall \boldsymbol{\alpha}$, $\boldsymbol{\beta} \in \mathcal{W}$; and
2. $k\boldsymbol{\alpha} \in \mathcal{W}$ $\forall \boldsymbol{\alpha} \in \mathcal{W}$, $k \in \mathcal{F}$,

then $\mathcal{W}$ is itself a linear space over the field $\mathcal{F}$. $\mathcal{W}$ is referred to as a linear subspace, or shortly subspace, of $\mathcal{V}$.

Since subspace is a linear space, the concepts of basis, dimension, representation and so on can be introduced. The details are clear and omitted. Since there may not be more linearly independent vectors in $\mathcal{W}$ than in $\mathcal{V}$, the dimension of a subspace $\mathcal{W}$ cannot exceed that of the whole space $\mathcal{V}$, that is, $\dim \mathcal{W} \leq \dim \mathcal{V}$.

1.4 Linear subspaces

- An example of linear subspaces

Let $A \in \mathcal{R}^{m\times n}$, then a solution to the homogeneous equation $A\boldsymbol{x} = \mathbf{0}$ is an n-dimensional column vector. The kernel of A is denoted by $\mathcal{N}(A)$ and referred to as the set of all solutions to $A\boldsymbol{x} = \mathbf{0}$. It can be readily checked that, for any $\boldsymbol{x}_1$, $\boldsymbol{x}_2 \in \mathcal{N}(A)$ and α_1, $\alpha_2 \in \mathcal{R}$, $\alpha_1\boldsymbol{x}_1 + \alpha_2\boldsymbol{x}_2 \in \mathcal{N}(A)$. Hence, $\mathcal{N}(A)$ is itself a linear space under the operations defined in $\mathcal{R}^n$. However, it is clear that $\mathcal{N}(A)$ is a proper subset of $\mathcal{R}^n$ as long as $A \neq 0$. Hence, $\mathcal{N}(A)$ is referred to as a linear subspace of $\mathcal{R}^n$.

1.4 Linear subspaces

- Span of vectors and related linear subspaces properties.

Let $\alpha_1, \alpha_2, \ldots, \alpha_s$ be a set of vectors of $\mathcal{V}$, the linear span of $\alpha_1, \alpha_2, \ldots, \alpha_s$ is referred to as the set of all possible linear representations of these vectors

$$\text{span}\{\alpha_1, \alpha_2, \ldots, \alpha_s\} = \{\alpha : \alpha = k_1\alpha_1 + k_2\alpha_2 + \cdots + k_s\alpha_s,\ k_i \in \mathcal{F}\}$$

Theorem 1.5 $\text{span}\{\alpha_1, \alpha_2, \ldots, \alpha_s\}$ is a subspace of $\mathcal{V}$.

1.4 Linear subspaces

Two theorems

Theorem 1.6 Every subspace $\mathcal{W} \subset \mathcal{V}$ is generated by some vectors $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_s$ in $\mathcal{V}$.

Theorem 1.7 For a given set of vectors $\{\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_s\} \subset \mathcal{V}$,

$$\dim \operatorname{span}\{\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_s\} = \operatorname{rank}\{\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_s\},$$

where $\operatorname{rank}\{\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_s\}$ is the number of vectors in any maximal linearly independent set of $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_s$. Any maximal linearly independent set of the generated space $\operatorname{span}\{\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_s\}$ can be its basis.

1.4 Linear subspaces

- Operations on linear subspaces (sum and intersection)

Let $\mathcal{U}$ and $\mathcal{V}$ be subspaces of $\mathcal{V}$. The intersect of $\mathcal{U}$ and $\mathcal{V}$ is denoted $\mathcal{U} \cap \mathcal{V}$ and referred to as the set

$$\mathcal{U} \cap \mathcal{V} = \{\boldsymbol{\alpha} \ : \ \boldsymbol{\alpha} \in \mathcal{U} \text{ and } \boldsymbol{\alpha} \in \mathcal{V}\}$$

It can be readily verified that $\mathcal{U} \cap \mathcal{V}$ is a subspace of $\mathcal{V}$. The sum of $\mathcal{U}$ and $\mathcal{V}$ is denoted $\mathcal{U} + \mathcal{V}$ and referred to as the set

$$\mathcal{U} + \mathcal{V} = \{\, \boldsymbol{\alpha} = \boldsymbol{\alpha}_1 + \boldsymbol{\alpha}_2 \ : \ \boldsymbol{\alpha}_1 \in \mathcal{U} \text{ and } \boldsymbol{\alpha}_2 \in \mathcal{V} \,\}$$

It can be checked that $\mathcal{U} + \mathcal{V}$ is a subspace of $\mathcal{V}$.

1.4 Linear subspaces

- We can define the union of subsets of V, however it is not a subspace.

The union of the subsets $\mathcal{U}$ and $\mathcal{V}$ is defined as

$$\mathcal{U} \cup \mathcal{V} \stackrel{\triangle}{=} \{ \boldsymbol{\alpha} \ : \ \boldsymbol{\alpha} \in \mathcal{U} \text{ or } \boldsymbol{\alpha} \in \mathcal{V} \}$$

$\mathcal{U} \cup \mathcal{V}$ is in general not a subspace. To show this, it suffices to find a counterexample.

1.4 Linear subspaces

- The union of subsets of V is not a subspace - an example

Let $\mathcal{V} = \mathcal{R}^2$, $\mathcal{U} = \text{span}\{\epsilon_1\}$ $\mathcal{V} = \text{span}\{\epsilon_2\}$ with $\epsilon_1 = [1 \quad 0]^T$ and $\epsilon_2 = [0 \quad 1]^T$. Then

$$\mathcal{U} \cup \mathcal{V} = \left\{ \boldsymbol{\alpha} \ : \ \boldsymbol{\alpha} = \begin{bmatrix} a_1 \\ 0 \end{bmatrix} \text{ or } \boldsymbol{\alpha} = \begin{bmatrix} 0 \\ a_2 \end{bmatrix} \right\}.$$

However,

$$\epsilon_1 \in \mathcal{U} \subset \mathcal{U} \cup \mathcal{V}, \quad \epsilon_2 \in \mathcal{V} \subset \mathcal{U} \cup \mathcal{V}$$

but

$$\epsilon_1 + \epsilon_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \notin \mathcal{U} \cup \mathcal{V}$$

The addition is not closed on the subset $\mathcal{U} \cup \mathcal{V}$, it is thus not a subspace.

1.4 Linear subspaces

- Two theorems

Theorem 1.8 Let

$$\begin{aligned} \mathcal{U} &= \text{span}\{\alpha_1, \alpha_2, \ldots, \alpha_s\}, \\ \mathcal{V} &= \text{span}\{\beta_1, \beta_2, \ldots, \beta_t\}, \end{aligned}$$

then

$$\mathcal{U}+\mathcal{V} = \text{span}\{\alpha_1, \alpha_2, \ldots, \alpha_s, \beta_1, \beta_2, \ldots, \beta_t\}$$

Theorem 1.9 Let $\mathcal{U}$ and $\mathcal{V}$ be subspaces of $\mathcal{V}$, then

$$\dim(\mathcal{U}+\mathcal{V}) = \dim\mathcal{U} + \dim\mathcal{V} - \dim(\mathcal{U}\cap\mathcal{V})$$

1.4 Linear subspaces

- Direct sum

Definition 1.19 Let $\mathcal{T}$ and $\mathcal{U}$ be subspaces of $\mathcal{V}$. If $\mathcal{T} \cap \mathcal{U} = \{\mathbf{0}\}$, then $\mathcal{T} + \mathcal{U}$ is direct sum. Direct sum is denoted $\mathcal{T} \dot{+} \mathcal{U}$.

Theorem 1.10 Let $\mathcal{T}$ and $\mathcal{U}$ be subspaces of $\mathcal{V}$. The following statements are equivalent:

1. $\mathcal{T} + \mathcal{U}$ is direct sum;
2. $\dim(\mathcal{T} + \mathcal{U}) = \dim \mathcal{T} + \dim \mathcal{U}$;
3. If $\begin{bmatrix} \boldsymbol{\alpha}_1 & \boldsymbol{\alpha}_2 & \dots & \boldsymbol{\alpha}_{n_1} \end{bmatrix}$ is a basis of $\mathcal{T}$, $\begin{bmatrix} \boldsymbol{\beta}_1 & \boldsymbol{\beta}_2 & \dots & \boldsymbol{\beta}_{n_2} \end{bmatrix}$ is a basis of $\mathcal{U}$, then

$$\begin{bmatrix} \boldsymbol{\alpha}_1 & \boldsymbol{\alpha}_2 & \dots & \boldsymbol{\alpha}_{n_1} & \boldsymbol{\beta}_1 & \boldsymbol{\beta}_2 & \dots & \boldsymbol{\beta}_{n_2} \end{bmatrix}$$

is a basis of $\mathcal{T} + \mathcal{U}$.

1.4 Linear subspaces

- Complementary subspace

Definition 1.20 Let $\mathcal{S}$, $\mathcal{T}$, $\mathcal{U}$ be subspaces of $\mathcal{V}$ and $\mathcal{U} = \mathcal{S} \dot{+} \mathcal{T}$, then $\mathcal{S} \dot{+} \mathcal{T}$ is referred to as a direct sum decomposition of $\mathcal{U}$. Especially, if

$$\mathcal{U} = \mathcal{V} = \mathcal{S} \dot{+} \mathcal{T}$$

then $\mathcal{S}$ and $\mathcal{T}$ are a pair of complementary subspaces with respect to $\mathcal{V}$. $\mathcal{S}$ ($\mathcal{T}$) is called the complementary subspace of $\mathcal{T}$ ($\mathcal{S}$) with respect to $\mathcal{V}$.

Theorem 1.11 For any subspace $\mathcal{T}$ of $\mathcal{U}$, there exists subspace $\mathcal{S} \subset \mathcal{U}$ such that

$$\mathcal{U} = \mathcal{S} \dot{+} \mathcal{T}$$

1.4 Linear subspaces

- An remark on complementary subspace

Given a subspace $\mathcal{T}$, its complementary subspace with respect to the whole space is not unique and depends on the non-unique extension of the basis of $\mathcal{T}$ to one of the whole space.

Exp: Let

$$\boldsymbol{\alpha}_1 = [1 \quad 0 \quad 0]^{\mathsf{T}}, \quad \boldsymbol{\alpha}_2 = [0 \quad 1 \quad 0]^{\mathsf{T}}$$

Then $\mathcal{T} = \text{span}\{\boldsymbol{\alpha}_1 \ , \ \boldsymbol{\alpha}_2\}$ is a subspace of $\mathcal{R}^3$. It is clear that both $\boldsymbol{\alpha}_3 = [0 \quad 0 \quad 1]^{\mathsf{T}}$ and $\boldsymbol{\alpha}_4 = [0 \quad 1 \quad 1]^{\mathsf{T}}$ can serve as an extension of $\boldsymbol{\alpha}_1$ and $\boldsymbol{\alpha}_2$ to a basis for $\mathcal{R}^3$. Hence, both $\text{span}\{\boldsymbol{\alpha}_3\}$ and $\text{span}\{\boldsymbol{\alpha}_4\}$ are complementary subspaces of $\mathcal{T}$ with respect to $\mathcal{R}^3$.

1.5 Linear transformation

·Linear transformation ---function mapping elements in a linear space to another

Definition 1.21 Let $\mathcal{U}$ and $\mathcal{V}$ be linear spaces over a field $\mathcal{F}$, a mapping $\mathscr{A} : \mathcal{U} \mapsto \mathcal{V}$ is said to be a linear transformation from $\mathcal{U}$ to $\mathcal{V}$ if it is additive and homogeneous, that is, for any two elements $\boldsymbol{\alpha}_1$, $\boldsymbol{\alpha}_2$ in $\mathcal{U}$ and any scalar k in $\mathcal{F}$

$$\mathscr{A}(\boldsymbol{\alpha}_1 + \boldsymbol{\alpha}_2) = \mathscr{A}(\boldsymbol{\alpha}_1) + \mathscr{A}(\boldsymbol{\alpha}_2),$$
$$\mathscr{A}(k\boldsymbol{\alpha}_1) = k\mathscr{A}(\boldsymbol{\alpha}_1)$$

$\mathscr{A}(\boldsymbol{\alpha}) \in \mathcal{V}$ is called the image of $\boldsymbol{\alpha} \in \mathcal{U}$ under $\mathscr{A}$. The totality of all images of the elements in $\mathcal{U}$ is denoted

$$\mathscr{A}(\mathcal{U}) \triangleq \{\boldsymbol{\beta} : \boldsymbol{\beta} \in \mathcal{V} \text{ and } \boldsymbol{\beta} = \mathscr{A}(\boldsymbol{\alpha}) \text{ for some } \boldsymbol{\alpha} \in \mathcal{U}\}$$

and is referred to as the image (or the range) of $\mathcal{U}$ under $\mathscr{A}$. $\mathcal{U}$ is the domain of $\mathscr{A}$.

1.5 Linear transformation

- Identity transformation and zero transformation

Exp: A transformation $\mathscr{E} : \mathcal{V} \mapsto \mathcal{V}$ is defined as

$$\mathscr{E}(\boldsymbol{\alpha}) = \boldsymbol{\alpha} \in \mathcal{V} \ \forall \ \boldsymbol{\alpha} \in \mathcal{V}$$

It can be readily checked that $\mathscr{E}$ is a linear transformation. $\mathscr{E}$ is called the identity transformation.

Exp: Let $\mathscr{O} : \mathcal{U} \mapsto \mathcal{V}$ be defined as

$$\mathscr{O}(\boldsymbol{\alpha}) = \mathbf{0} \ \forall \ \boldsymbol{\alpha} \in \mathcal{U}$$

Then it can be readily checked that $\mathscr{O}$ is a linear transformation. Since $\mathscr{O}$ transforms all elements in $\mathcal{U}$ identically into $\mathbf{0} \in \mathcal{V}$, it is called the zero transformation.

1.5 Linear transformation

- Properties of linear transformation

  A linear transformation has the following properties.

  1. $\mathscr{A}(\mathbf{0}) = \mathbf{0}$;

  2. $\mathscr{A}\left(\sum_{i=1}^{s} k_i \boldsymbol{\alpha}_i\right) = \sum_{i=1}^{s} k_i \mathscr{A}(\boldsymbol{\alpha}_i)$;

  3. If $\boldsymbol{\alpha}_1, \boldsymbol{\alpha}_2, \ldots, \boldsymbol{\alpha}_s \in \mathcal{V}$ are linearly dependent, then the images $\mathscr{A}(\boldsymbol{\alpha}_1), \mathscr{A}(\boldsymbol{\alpha}_2), \ldots, \mathscr{A}(\boldsymbol{\alpha}_s)$ are also linearly dependent.

1.5 Linear transformation

- An example of linear transformation

Exp: Let

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

be an $m \times n$ real matrix. It can be readily checked that $\mathscr{A} : \mathcal{R}^n \mapsto \mathcal{R}^m$ defined as

$$\mathscr{A}(\boldsymbol{\alpha}) = A\boldsymbol{\alpha} \in \mathcal{R}^m \ \forall \ \boldsymbol{\alpha} \in \mathcal{R}^n$$

is a linear transformation from $\mathcal{R}^n$ to $\mathcal{R}^m$.

1.5 Linear transformation

- Remark on linear transformation

If the vectors $\alpha_1, \alpha_2, \ldots, \alpha_s \in \mathcal{U}$ are linearly dependent, the the images $\mathscr{A}(\alpha_1), \mathscr{A}(\alpha_2), \ldots, \mathscr{A}(\alpha_s)$ must be linearly dependent. However, the statement cannot be made in the opposite direction. Even though the vectors $\alpha_1, \alpha_2, \ldots, \alpha_s \in \mathcal{U}$ are linearly independent, their images $\mathscr{A}(\alpha_1), \mathscr{A}(\alpha_2), \ldots, \mathscr{A}(\alpha_s)$ may be linearly dependent, -see e.g. the zero transformation. is a linear transformation from $\mathcal{R}^n$ to $\mathcal{R}^m$.

1.5 Linear transformation

• Matrix representation of a linear transformation

For notational simplicity, we shall denote by $\mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$ the set of all linear transformations from the linear space $\mathcal{U}$ to the linear space $\mathcal{V}$ over the field $\mathcal{F}$. Let $[\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n]$ be a basis of $\mathcal{U}$, and $[\boldsymbol{\beta}_1 \quad \boldsymbol{\beta}_2 \quad \dots \quad \boldsymbol{\beta}_m]$ be a basis of $\mathcal{V}$, $\mathscr{A} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$ be a linear transformation, then the image of a basis vector in $\mathcal{U}$ can be represented as

$$\mathscr{A}(\boldsymbol{\alpha}_j) = \sum_{i=1}^{m} a_{ij}\boldsymbol{\beta}_i = [\boldsymbol{\beta}_1 \quad \boldsymbol{\beta}_2 \quad \dots \quad \boldsymbol{\beta}_m] \underbrace{\begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}}_{\triangleq \boldsymbol{a}_j}$$

the totality of the images of all the basis vectors can be written as

$$\begin{aligned} \mathscr{A}([\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n]) &\triangleq [\mathscr{A}(\boldsymbol{\alpha}_1) \quad \mathscr{A}(\boldsymbol{\alpha}_2) \quad \dots \quad \mathscr{A}(\boldsymbol{\alpha}_n)] \\ &= \left[\sum_{i=1}^{m} a_{i1}\boldsymbol{\beta}_i \quad \sum_{i=1}^{m} a_{i2}\boldsymbol{\beta}_i \quad \dots \quad \sum_{i=1}^{m} a_{in}\boldsymbol{\beta}_i\right] \\ &= [\boldsymbol{\beta}_1 \quad \boldsymbol{\beta}_2 \quad \dots \quad \boldsymbol{\beta}_m] \underbrace{[\boldsymbol{a}_1 \quad \boldsymbol{a}_2 \quad \dots \quad \boldsymbol{a}_n]}_{\triangleq A} \end{aligned}$$

1.5 Linear transformation

- Matrix representation of a linear transformation

A is referred to as the matrix representation of the linear transformation $\mathscr{A} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$ with respect to the bases $[\alpha_1 \quad \alpha_2 \quad \dots \quad \alpha_n]$ and $[\beta_1 \quad \beta_2 \quad \dots \quad \beta_m]$.

A direct use of the matrix representation is to evaluate the representation of the image $\mathscr{A}(\alpha)$ in relation to the representation of any $\alpha \in \mathcal{U}$.

- Matrix representation gives the relation of an element in a linear space and its images in another space.

1.5 Linear transformation

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Let $\boldsymbol{\alpha} \in \mathcal{U}$, then

$$\boldsymbol{\alpha} = [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n] \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}_{=\boldsymbol{x}}$$

- Coordinates transformation

The image $\mathscr{A}(\boldsymbol{\alpha}) \in \mathcal{V}$ can be represented as

$$\mathscr{A}(\boldsymbol{\alpha}) = [\boldsymbol{\beta}_1 \quad \boldsymbol{\beta}_2 \quad \dots \quad \boldsymbol{\beta}_m] \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}}_{=\boldsymbol{y}}$$

From

$$\begin{aligned} \mathscr{A}(\boldsymbol{\alpha}) &= \mathscr{A}([\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n]\boldsymbol{x}) \\ &= \mathscr{A}(x_1\boldsymbol{\alpha}_1 + x_2\boldsymbol{\alpha}_2 + \cdots + x_n\boldsymbol{\alpha}_n) \\ &= \mathscr{A}(\boldsymbol{\alpha}_1)x_1 + \mathscr{A}(\boldsymbol{\alpha}_2)x_2 + \cdots + \mathscr{A}(\boldsymbol{\alpha}_n)x_n \\ &= [\mathscr{A}(\boldsymbol{\alpha}_1) \quad \mathscr{A}(\boldsymbol{\alpha}_2) \quad \dots \quad \mathscr{A}(\boldsymbol{\alpha}_n)]\,\boldsymbol{x} \\ &= [\boldsymbol{\beta}_1 \quad \boldsymbol{\beta}_2 \quad \dots \quad \boldsymbol{\beta}_m]\,A\boldsymbol{x} \end{aligned}$$

and the uniqueness of the representation with respect to a given fixed basis, it follows that

$$\boldsymbol{y} = A\boldsymbol{x}$$

1.5 Linear transformation

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Let $\boldsymbol{\alpha} \in \mathcal{U}$, then

$$\boldsymbol{\alpha} = [\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n] \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}_{=\boldsymbol{x}}$$

The image $\mathscr{A}(\boldsymbol{\alpha}) \in \mathcal{V}$ can be represented as

$$\mathscr{A}(\boldsymbol{\alpha}) = [\boldsymbol{\beta}_1 \quad \boldsymbol{\beta}_2 \quad \dots \quad \boldsymbol{\beta}_m] \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{bmatrix}}_{=\boldsymbol{y}}$$

From

$$\begin{aligned} \mathscr{A}(\boldsymbol{\alpha}) &= \mathscr{A}([\boldsymbol{\alpha}_1 \quad \boldsymbol{\alpha}_2 \quad \dots \quad \boldsymbol{\alpha}_n]\boldsymbol{x}) \\ &= \mathscr{A}(x_1\boldsymbol{\alpha}_1 + x_2\boldsymbol{\alpha}_2 + \cdots + x_n\boldsymbol{\alpha}_n) \\ &= \mathscr{A}(\boldsymbol{\alpha}_1)x_1 + \mathscr{A}(\boldsymbol{\alpha}_2)x_2 + \cdots + \mathscr{A}(\boldsymbol{\alpha}_n)x_n \\ &= [\mathscr{A}(\boldsymbol{\alpha}_1) \quad \mathscr{A}(\boldsymbol{\alpha}_2) \quad \dots \quad \mathscr{A}(\boldsymbol{\alpha}_n)]\,\boldsymbol{x} \\ &= [\boldsymbol{\beta}_1 \quad \boldsymbol{\beta}_2 \quad \dots \quad \boldsymbol{\beta}_m]\,A\boldsymbol{x} \end{aligned}$$

and the uniqueness of the representation with respect to a given fixed basis, it follows that

$$\boldsymbol{y} = A\boldsymbol{x}$$